Results link : updated results

Code file link(to run these codes we need to pass log-returns based on two assets or 4 assets and we can change the stepsize etc inside the code and can run the files in GPU) :

Code files when threshold function is updated :

for horizon 10 and with KDE
for horizon 10 and without kde
for horizon 1 and without kde
for horizon 1 and with kde

Log returns link : log_4returns
log_returns

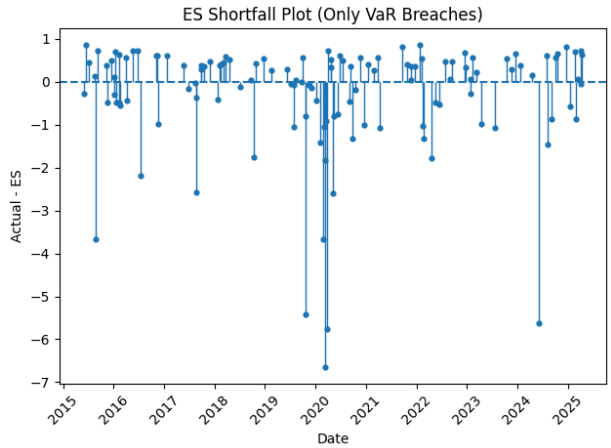
If we see the results(sheet 1) we can observe that always, with kde model has less mean FZ0 score than no kde model so with kde model is better as our primary evaluation factor is FZ0 score. with no kde model , all backtests gave satisfactory results but in kde model for the case step size 10 , horizon 10 , 4 assets, dq test with lags 4 has p value less than 0.05. So i have increased the simulations to 20k from 10k as you have mentioned and the result is not changing so it is not because of noise .we could also observe that there is not much difference with copula selection based on last 50 or 25 or 100 points . In almost every case fallback occurs less than 15 percent only .

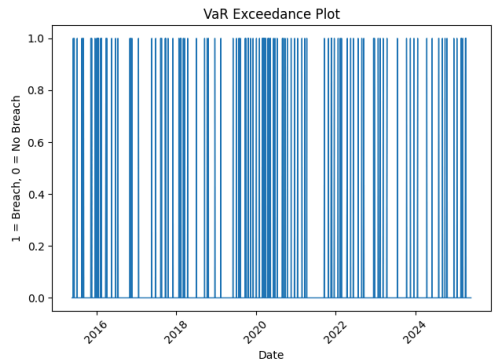
After that above update :

Code link : for h1 and with kde, updated threshold,pit.ipynb
for h10 and with kde, updated threshold,pit.ipynb
for h10 and without kde, updated threshold,pit.ipynb
for h1 and without kde, updated threshold,pit.ipynb

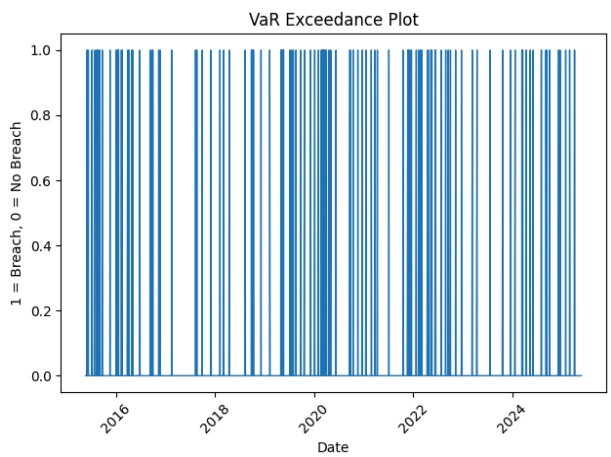
While running above files i have gone through entire code once and i realised one mistake that is, during pit construction i have not normalised KDE , this might be the reason for jump at threshold and that is the reason we are using blending . but now i have normalised the KDE and removed the blending part . I have started running these files after changes, i am posting results in the same excel sheet(sheet 2). now after updating kde normalization where blending is also removed , we can observe that with kde and with no kde model is giving almost same FZo score and also i have checked ks test results(these are also satisfactory) . and also before normalization ,kde model is failing in one test only (for dq test with 4 lags) , in this test implementation degrees of freedom are mentioned wrong , plus 1 should be there so with this change, that dq test error got removed(it is giving satisfactory results after updating it) . we can see that fallback is applied less than 15 percent so the difference comes between with kde and with no kde in less than 15 percent cases only which might be reason for not having much difference. Here after this update if we consider that small gap between with Kde and with no kde then with no kde is given less FZ0 score in all cases .

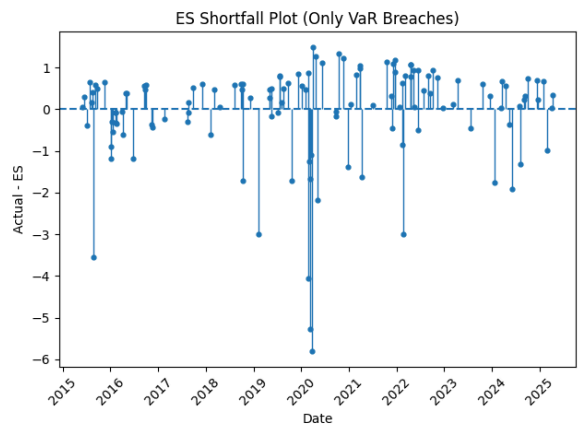
These are the tables before updating KDE normalization :
h=1 s=1 assets = 2 with kde



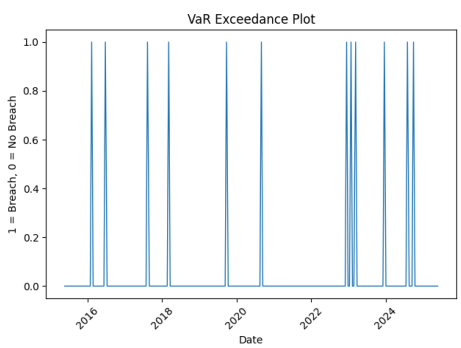


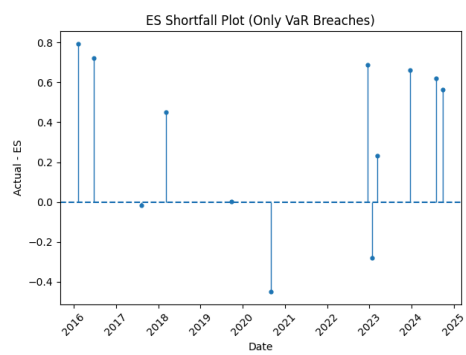
h=1 s=1 assets = 4



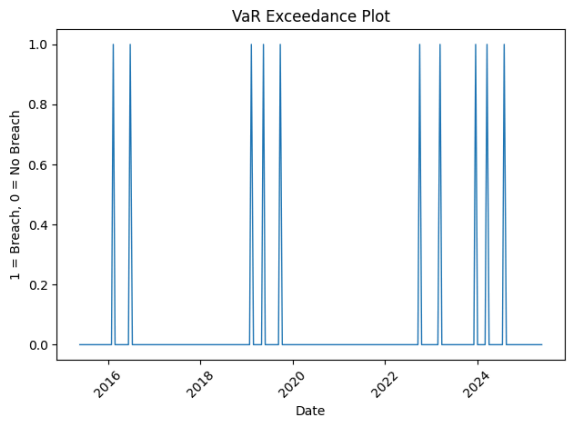


h=1 s=10 assets = 2

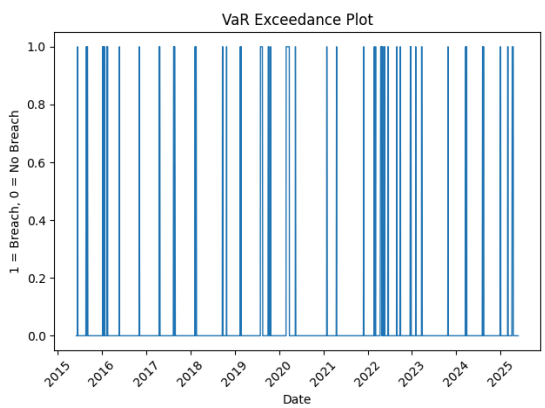


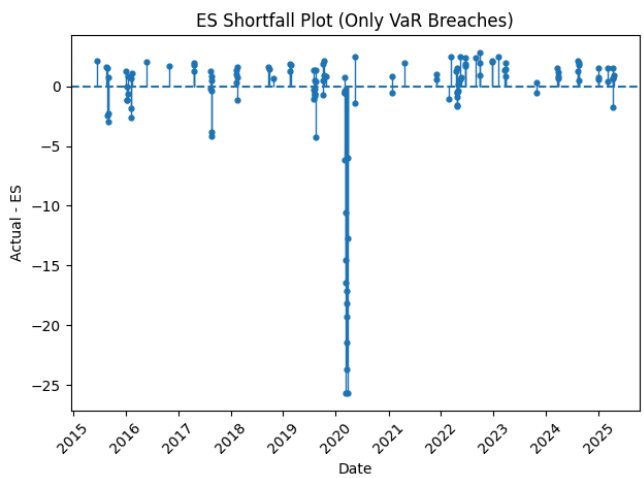


h=1 s=10 assets = 4

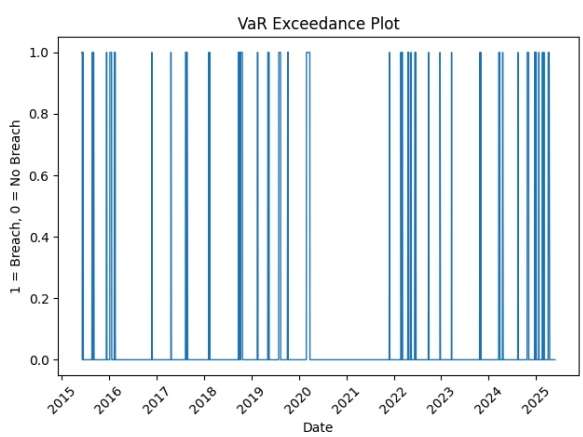


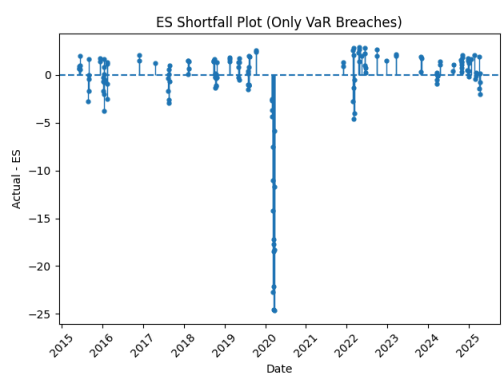
h=10 s=1 assets = 2



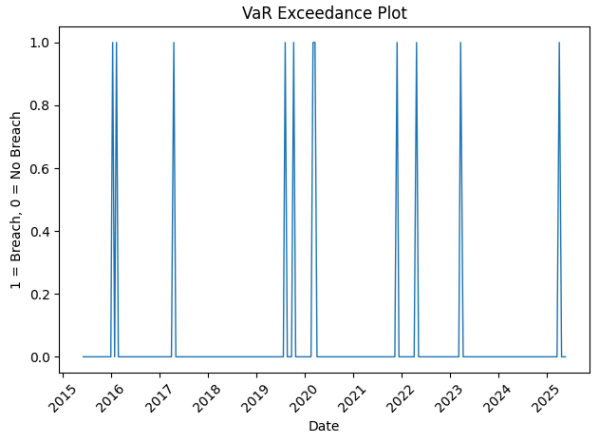


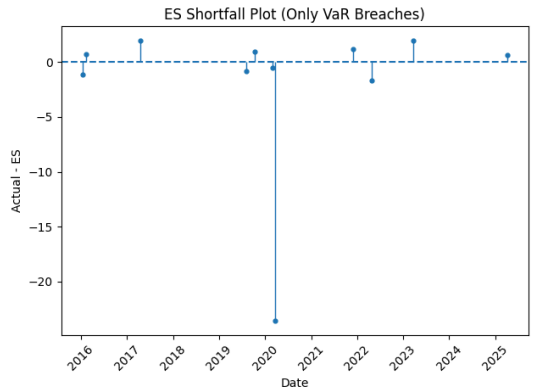
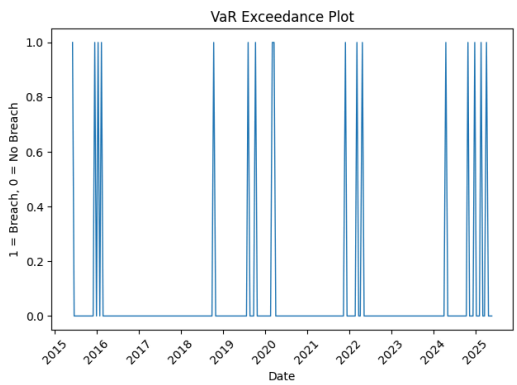
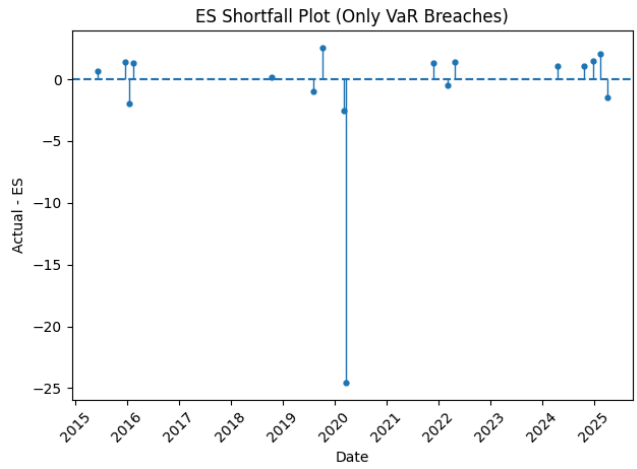
h=10 s=1 assets = 4





h=10 s=10 assets = 2





h=10 s=10 assets = 4